Chapter 18 - Antiderivatives

#Antiderivative (Indefinite Integral) and Antidifferentiation (Integration)

Antiderivative (or Indefinite Integral) is the inverse function of the derivative which is the instantaneous rate of change of 'y' w.r.t. 'x'. Antidifferentiation (or Integration) is the process of finding the antiderivative of a function. The main idea of Integration is to reverse the Differentiation.

Notations of Antiderivative: $\int f(x)\,dx, F(x)$

The antiderivative of $f(x)$ w.r.t. 'x' is $F(x)$ means that the derivative of $F(x)$ w.r.t. 'x' is $f(x)$.

Symbolically, $\int f(x)\,dx = F(x) \Rightarrow \frac{d}{dx}F(x) = f(x)$

#Basic Properties of Indefinite Integral

1. $\int k\,f(x)\,dx = k\int f(x)\,dx$
2. $\int [f(x) \pm g(x)]\,dx = \int f(x)\,dx \pm \int g(x)\,dx$

#Integration as reverse of Differentiation

Integrals involving algebraic functions:

1. $\int x^n \, dx = \frac{x^{n+1}}{n+1} + c, (n \neq -1)$
2. $\int \frac{1}{x} \, dx = \log x + c$
3. $\int k \, dx = kx + c$
4. $\int (ax + b)^n \, dx = \frac{(ax+b)^{n+1}}{a(n+1)} + c, (n \neq -1)$
5. $\int \frac{1}{ax+b} \, dx = \frac{\log(ax+b)}{a} + c$

Integrals involving trigonometric functions:

1. $\int \cos x \, dx = \sin x + c$
2. $\int \sin x \, dx = -\cos x + c$
3. $\int \sec^2 x \, dx = \tan x + c$
4. $\int \operatorname{cosec}^2 x \, dx = -\cot x + c$
5. $\int \sec x \tan x \, dx = \sec x + c$
6. $\int \operatorname{cosec} x \cot x \, dx = -\operatorname{cosec} x + c$

Integral of exponential function

$\int e^x \, dx = e^x + c$

#Composite Function Rule of Integration

$\int f(ax + b) \, dx = \frac{F(ax+b)}{a} + c$

Note: This can only be applied when the inner function is a linear function.

#Integration by Substitution Method

$\int f'(x)\, g[f(x)]\, dx$

Substitute $y = f(x) \Rightarrow dy = f'(x)\, dx$

Now,

$\int g[f(x)]\, f'(x)\, dx = \int g(y)\, dy = G(y) + c$

Integration of trigonometric functions

$\int sin\, x\, dx = - cos\, x + c$

$\int cos\, x\, dx = sin\, x + c$

$$\int \tan x\ dx = \begin{cases} -\log \cos x + c \\ \log \sec x + c \end{cases}$$

$$\int \cot x\ dx = \begin{cases} \log \sin x + c \\ -\log \csc x + c \end{cases}$$

$$\int \sec x\ dx = \begin{cases} -\log(\sec x - \tan x) + c \\ \log(\sec x + \tan x) + c \end{cases}$$

$$\int \csc x\ dx = \begin{cases} \log(\csc x - \cot x) + c \\ -\log(\csc x + \cot x) + c \end{cases}$$

Integration involving exponential function

1. $\int f'(x)\, e^{f(x)} dx$

   Substitute $y = f(x) \Rightarrow dy = f'(x)\, dx$

   Now, $\int f'(x)\, e^{f(x)} dx = \int e^{y} dy = e^{y} + c$

2. $\int \frac{f'(x)}{f(x)}\, dx$

   Substitute $y = f(x) \Rightarrow dy = f'(x)\, dx$

   Now, $\int \frac{f'(x)}{f(x)}\, dx = \int \frac{dy}{y} = log\, y + c$

#Integration by Trigonometric Substitution

1. $a^2 - x^2 \xrightarrow{x = a\sin\theta} a^2\cos^2\theta$
2. $a^2 + x^2 \xrightarrow{x = a\tan\theta} a^2\sec^2\theta$
3. $x^2 - a^2 \xrightarrow{x = a\sec\theta} a^2\tan^2\theta$

Special Case: $\displaystyle \int \frac{dx}{\sqrt{x^2 \pm a^2}} \xrightarrow{y = x + \sqrt{x^2 \pm a^2}} \int \frac{dy}{y}$

#Product Rule of Integration (Integration by Parts)

$$\int uv\,dx = u\int v\,dx - \int \frac{du}{dx}\int v\,dx\,dx$$

or,

$$\int f(x)\,g(x)\,dx = f(x)\int g(x)\,dx - \int \frac{d\,f(x)}{dx}\int g(x)\,dx\,dx$$

We use either of the following rule to choose the 1st and 2nd function in the product:

ILATE Rule

For easier work, we use the following priority order to choose the 1st function 'u':

I → Inverse Trigonometric

L → Logarithmic

A → Algebraic

T → Trigonometric

E → Exponential

DI Rule

We can also choose the functions such that

D → Derivative of 1st function 'u' should be easier to find

I → Integral of 2nd function 'v' should be easier to find

#Definite Integrals

Fundamental Theorem of Integral Calculus

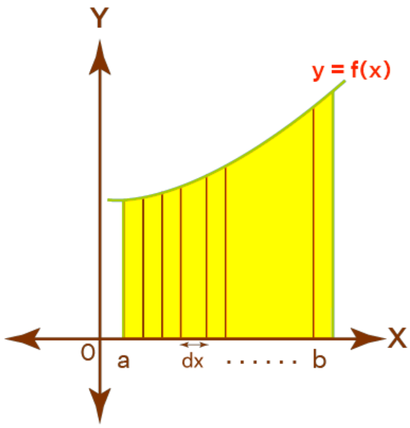
The definite integral of $f(x)$ w.r.t. 'x' from a to b means $[F(x)]_a^b$ that is $F(b) - F(a)$.

Symbolically, $\int_a^b f(x)\, dx = [F(x)]_a^b = F(b) - F(a)$.

Change of Limits

When we change the variable by substitution, it's also essential to change the limits as well.

#Definite Integral as Area under curve



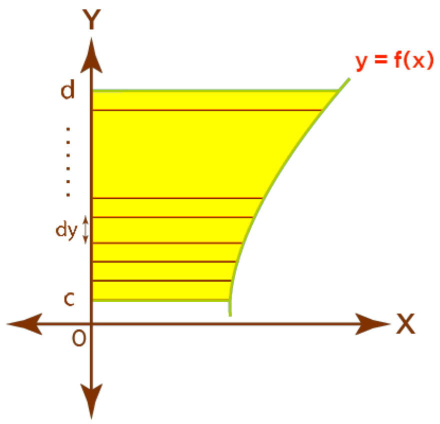
The notation $\int_a^b f(x)\, dx$ gives the sum of all $f(x)\, dx$ in the interval $[a, b]$ where dx is an infinitesimal change in x.

So the definite integral gives the area under the curve and over the bounding axis in the given interval.

Hence, Area bounded by x-axis on $[a, b]$ is $A = \int_a^b f(x)\, dx$.

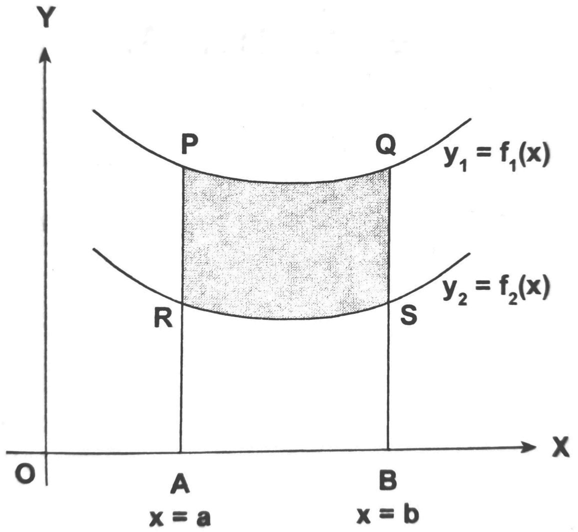
Corollary

We can also do the same about y-axis



Area bounded by y-axis on $[c, d]$ is $A = \int_c^d x\, dy$

Area between two curves



When y_1 lies above y_2 then the area between the two curves bounded by X-axis in the interval $[a, b]$ is

$$A = \int_a^b y_1\,dx - \int_a^b y_2\,dx$$

Since the area is never negative, so we take the positive values of both definite integrals

$$A = \left|\int_a^b y_1\,dx\right| - \left|\int_a^b y_2\,dx\right|$$

Note: The definite integral may be negative but the area under the curve is never negative.